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THE UNIVERSITY OF HONG KONG

COMP7705

MSc(CompSc) Project Progress 2 Report

An innovative AI-powered interview assessment tool for company

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1. Executive Summary

This reporting period delivered a substantial transition from a text-only AI interviewer to a multimodal interview system with integrated voice interaction and a virtual interviewer interface. The implementation is aligned with the planned milestone trajectory and has already initiated preparatory work for Task 4 (Advanced Features), particularly in multidimensional scoring and richer behavioral signal analysis.

2. Recap of Previous Phase (Brief)

In the previous phase (Phase 1 & 2), the core text-based interview pipeline was stabilized, including:

1. AI-driven interview flow with follow-up logic,
2. API integration for LLM response generation,
3. automatic post-interview evaluation and scoring, and
4. persistent data management via SQLite.

3.Milestones

Tasks	Estimated completion time	Estimated number of learning hours
~~1. Research & Planning~~ ~~- Review AI Hiring PDF and related literature~~ ~~- Define objectives, scope, and stream (Multimedia Computing)~~ ~~- Install Python, Streamlit, OpenAI/Groq API - Draft and submit Detailed Project Proposal~~ ~~- Create and initialize project webpage on school account~~	~~March 10, 2026~~	~~60 hours~~

~~2. Basic Interface & Question Generation~~ - ~~Build simple chat interface using Streamlit~~ - ~~Implement basic question generation with LLM API (prompt engineering)~~ - ~~Test with sample job descriptions~~ - ~~Update project webpage with screenshots and progress~~	~~April 7, 2026~~	~~50 hours~~
~~3. Multi-turn Conversation & Initial Scoring~~ - ~~Add conversation history management for multi-turn interviews~~ - ~~Develop initial automated scoring (relevance, technical depth)~~ - ~~Conduct preliminary tests with dummy responses~~ - ~~Update webpage with demo video/screenshots~~	~~May 5, 2026 (Progress Update 2)~~	~~45 hours~~

4. Advanced Features & Interim Report - ~~Implement full scoring dimensions (clarity, value alignment)~~ - ~~Add intelligent follow-up probes~~ - ~~Generate basic reports and recommendations~~ - Prepare and deliver Interim Report & Presentation	June 1, 2026 (Interim Report & Presentation)	50 hours
5. Testing, Refinement & Bias Check - Full system end-to-end testing with 20–30 dummy cases - Check for bias and improve prompts - Fix bugs and optimize performance Update webpage with test results	June 16, 2026 (Progress Update 3)	70 hours
6. Final Polish & Oral Preparation - Refine UI and report format - Conduct final full tests and collect results - Prepare demo video and oral examination materials - Final webpage update	July 6, 2026 (Progress Update 4)	50 hours
7. Final Submission - Complete Project Report - Submit final webpage (July 13)	July 17, 2026 (Project Report)	30 hours

4. Current Implementation Progress

4.1 Voice Conversation Integration

The system now supports bidirectional voice interaction through a complete STT–LLM–TTS pipeline:

- Speech-to-Text (STT): Candidate audio is captured via an in-app recording component and transcribed using local Whisper inference **as shown in Figure 1**.
- Text-to-Speech (TTS): Interviewer responses are synthesized using edge-tts with HTTP chunk streaming to reduce perceived response latency **as shown in Figure 2**.
- Interface-level integration: The candidate interface now includes recording controls, transcription feedback, and synchronized spoken interviewer output.

A notable engineering contribution in this phase is improved Windows runtime reliability through ffmpeg-path handling and controlled fallback behavior for speech synthesis.

4.2 Virtual Interviewer Avatar

The project implemented a dedicated avatar rendering frontend and integrated it with the interview state machine:

- A session-based avatar state endpoint was introduced for interviewer utterance synchronization.
- A React/Vite avatar panel was connected to the interview flow and embedded into the Streamlit interface for single-page operation **as shown in Figure 3**.
- Multiple realism-performance strategies were tested:
  - static image with pseudo lip-motion (limited realism)
  - looped human video with speech-state synchronization (current practical baseline)
  - preliminary pause-aware playback logic (speech activity driven)

4.3 Advanced Scoring (Task 4 Preparation)

Task 4 has been partially initiated by extending the evaluation output structure beyond a single scalar score:

- Multi-criteria rubric scaffolding has been introduced (e.g., relevance/completeness, clarity, reasoning under follow-up).
- Structured score decomposition fields were added for downstream analytics and reporting **as shown in Figure 4**.
- The evaluation module now supports evidence-oriented fields (reasoning traces and interpretable breakdown candidates), preparing for full Task 4 dimensional scoring.

5. Technical Challenges and Solutions

5.1 Latency Under Multimodal Interaction

Challenge: End-to-end delay increased when combining LLM generation, STT, TTS, and avatar synchronization.

Solution:

- Token-level/streamed response rendering for reduced perceived waiting time.
- Chunked TTS streaming to start playback before full synthesis completion.
- Decoupled avatar state updates from core interview logic to avoid blocking the primary dialogue loop.

5.2 STT Robustness and Accuracy

Challenge: Recognition quality variability due to accent diversity, recording quality, and platform-specific ffmpeg dependency issues.

Solution:

- Adoption of local Whisper for controllable inference.
- Runtime ffmpeg compatibility handling for Windows.

- Explicit transcription visibility in UI for transparent verification and debugging.

5.3 Avatar Integration Cost vs. Realism

Challenge: High-realism talking-head generation is computationally expensive and typically not suitable for strict real-time interview interaction.

Solution:

- Offline-generation approaches were evaluated but not selected for the live path due to latency overhead.
- A pragmatic real-time visual strategy (looped interviewer video and synchronized speech states) was adopted as a middle ground between realism and responsiveness.

6. Next Steps and Future Work

In preparation for the Interim Report & Presentation milestone (Task 4), the following items are prioritized:

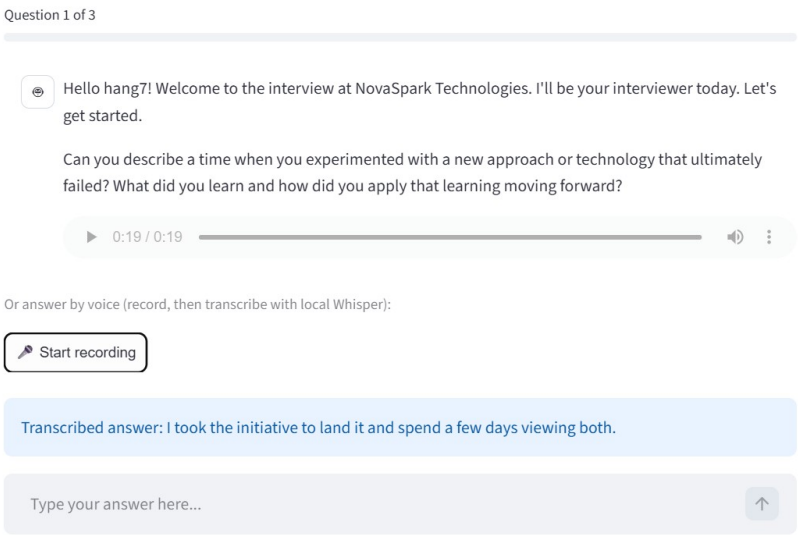
1. Full multidimensional scoring implementation
   Expand and validate complete scoring dimensions (e.g., clarity, value alignment, behavioral depth) with calibrated weighting.
2. Anti-deepfake and liveness safeguards
   Design camera-based liveness checks and identity consistency controls for interview integrity.
3. Facial expression signal integration
   Add optional expression-level behavioral indicators as supplementary evidence channels (not as standalone decision criteria).
4. Voice tone analysis
   Incorporate prosodic features (speech rate, pause profile, intonation stability) to complement content-based assessment.
5. Reliability hardening for deployment

Further stabilize browser audio policy handling, synchronization fault recovery, and session-level observability for demonstration robustness.

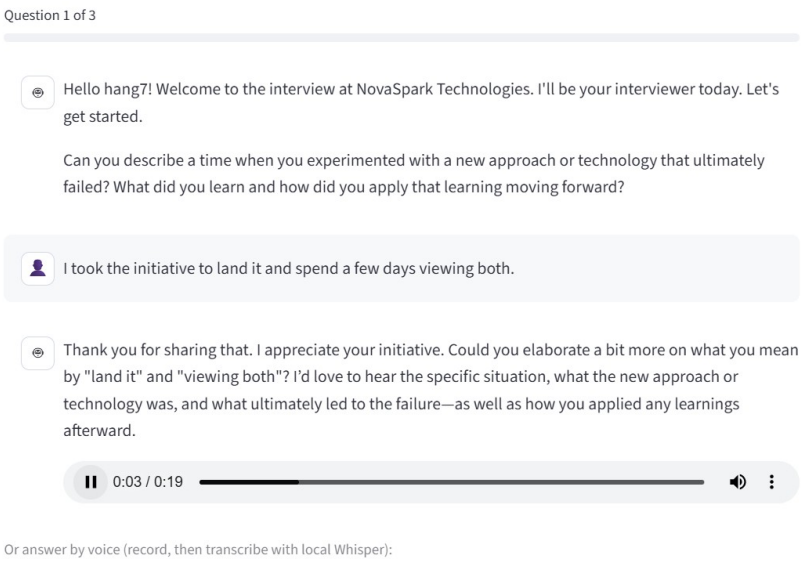
7.Project Website

The project webpage is live at https://wp2025.cs.hku.hk/msp25020/ and has been regularly updated with screenshots, progress updates, and live demo links.

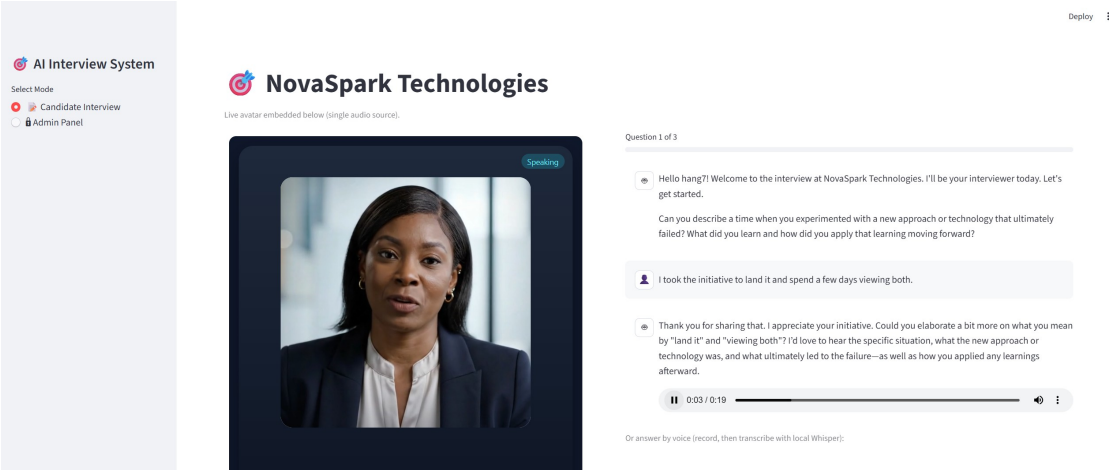
8.Figures



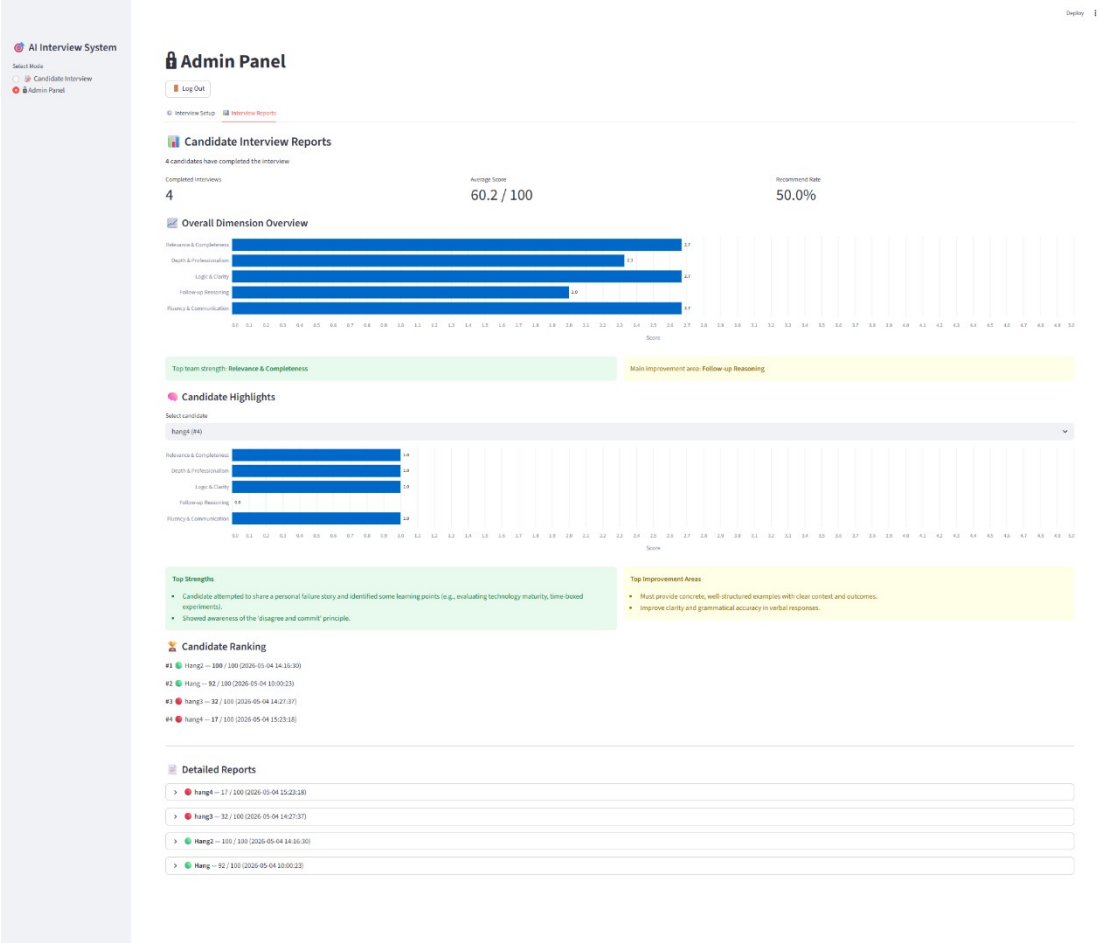
(**Figure 1.** Local Whisper transcription output shown in the interview workflow)



(**Figure 2.** Interviewer response with synchronized TTS streaming playback)



(**Figure 3.** Single-page integration of Streamlit interview UI and React avatar panel.)



(**Figure 4.** Admin-side report view with score and structured evaluation fields.)